

Milestone One – Detailed Documentation

This milestone covers two core phases of the project: Module 1 (Web Scraping & Raw Data Collection) and Module 2 (Data Cleaning & Normalization). These modules together form the foundation for all further analytics, ranking, and visualization.

Module 1: Web Scraping & Raw Data Collection

Methodology:

Data was collected from GlobalFirepower.com, which provides structured military, economic, and geographic metrics. A base page was scraped to retrieve global ranking and Power Index. Over 50 metric pages were then scraped using automated HTTP requests. Data was parsed using HTML selectors and merged using country names as keys. Throttling and validation ensured ethical scraping.

Libraries Used:

- 1 requests – HTTP communication
- 2 BeautifulSoup – HTML parsing
- 3 pandas – Data handling
- 4 re – URL and value extraction
- 5 time – Request control

Outputs:

- 1 military raw data.csv
- 2 military raw data.xlsx
- 3 military pivot data.csv

Module 2: Data Cleaning & Normalization

Raw pivoted data was cleaned by removing special characters and units. Values were converted to numeric format. Missing values were filled using median imputation. This produced a clean, standardized dataset ready for dashboards, analysis, and ranking.

Libraries Used:

- 1 pandas
- 2 numpy
- 3 re

Final Output:

military cleaned.csv

Milestone One Outcome

The milestone successfully delivered an automated data pipeline, raw and cleaned datasets, and a structured foundation for advanced military power analysis.